

WASP-104b

R-2571

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_170222 · created 2026-08-26T23:56:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457805.1714 +/- 0.00073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.044
Transit depth (Rp/Rs)^2	1.92 +/- 1.22 [%]
Orbital Inclination (inc)	82.34 +/- 2.88
Airmass coefficient 1 (a1)	2.01 +/- 1.21
Airmass coefficient 2 (a2)	-0.49 +/- 1.33
Scatter in the residuals of the lightcurve fit is	59.05 %
Best Comparison Star	#3 - [535, 316]
Optimal Aperture	2.88
Optimal Annulus	4.75
Transit Duration (day)	0.025 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.044 vs 0.1214 (deviation 0.26 σ)
Duration fit vs published	0.025 d vs 0.0742 d (deviation 2.13 σ)
Mid-time O-C	-2526.1 min
Depth SNR	2.2
Residual scatter	59.05 %

Light curve

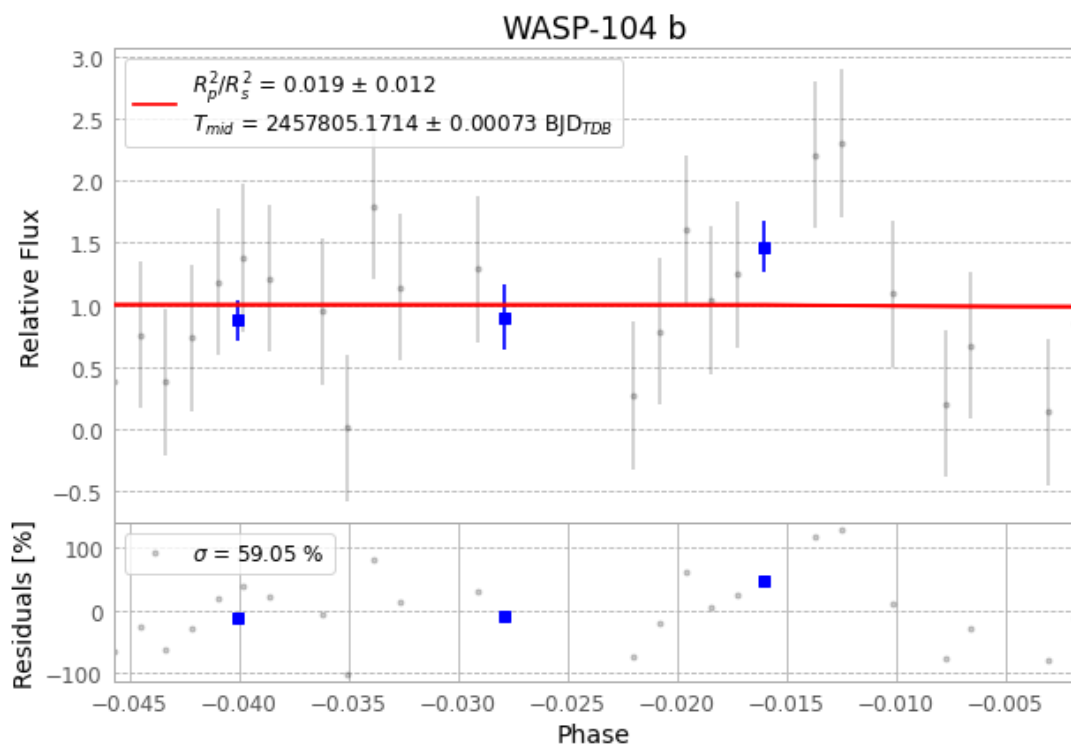


Figure 1 — FinalLightCurve_WASP-104 b_2017-02-22.png (SHA-256 9b2be5cd2239b8c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [166, 367], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1b453ca9d3bb8871...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

39 FITS frames (26 MB), WASP-104170222080912.FITS → WASP-104170222100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-170222.log (34bc19559d04...), FinalLightCurve_WASP-104 b_2017-02-22.png (9b2be5cd2239...), FinalLightCurve_WASP-104 b_2017-02-22.csv (2420c645bbad...)

Compute resources

Wall time 81.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:56Z.